MA1521 Final Exam Helpsheet

Thomas 15/e §11.3–11.5 · Ch. 13 · Ch. 14 | side 1: Q1–17 · side 2: Q18–28 | SF = sample-final question type

0 Single-variable (Q1–3)

- **L'Hôpital.** $\frac{0}{0}$ or $\frac{\infty}{\infty}$: $\lim \frac{f}{g} = \lim \frac{f'}{g'}$ — **re-check the form before each use**; repeat if still $0/0$. $0\cdot\infty$: write as $f/(1/g)$. $1^\infty, 0^0, \infty^0$: take ln first; $\lim\left(1+\frac{a}{x}\right)^{bx} = e^{ab}$.
- **Maclaurin shortcut** near 0: $e^x = 1 + x + x^2/2 + \ldots$, $\sin x = x - x^3/6$, $\cos x = 1 - x^2/2$, $\ln(1+x) = x - x^2/2$.

SF Q2 $\lim_{x\to 2} \frac{(x-2)(\ln x - \ln 2)}{x^2-5x+6+2\ln x - \ln 4}$: L'H twice $\Rightarrow \frac{1/x+2/x^2}{2-2/x^2} = \frac{1}{3/2} = \frac{2}{3}$.

- **Improper integrals.** Bad endpoint $\Rightarrow$ lim. $\int_0^1 x^n \ln x\,dx = -\frac{1}{(n+1)^2}$, $\int_0^1 x^n(\ln x)^2 dx = \frac{2}{(n+1)^3}$, $\int_0^\infty x^n e^{-ax}dx = \frac{n!}{a^{n+1}}$. Boundary terms: $\lim_{x\to 0^+} x^a(\ln x)^k = 0$ $(a>0)$, by L'Hôpital on $(\ln x)^k/x^{-a}$.

SF Q1 $\int_0^1 x(\ln x)^2 dx$: parts twice ($u=(\ln x)^2$, then $u=\ln x$) $\Rightarrow \left[\frac{x^2}{2}(\ln x)^2 - \frac{x^2}{2}\ln x + \frac{x^2}{4}\right]_{0^+}^1 = \frac{1}{4}$.

- **FTC.** $\frac{d}{dx}\int_{a(x)}^{b(x)} g(t)dt = g(b(x))b'(x) - g(a(x))a'(x)$.

SF Q3 $f = \int_{x-2}^{x^2-7} \frac{dt}{t^3+3}$: $f'(3) = \frac{2\cdot 3}{2^3+3} - \frac{1}{1^3+3} = \frac{6}{11} - \frac{1}{4} = \frac{13}{44}$.

- Parts: $\int u\,dv = uv - \int v\,du$ (LIATE). Sub: **change the limits**. $\int \ln x = x\ln x - x$.

11 Vectors, lines & planes (Q4)

- $\mathbf{u}\cdot\mathbf{v} = \sum u_i v_i = |\mathbf{u}||\mathbf{v}|\cos\theta$; $\perp \Leftrightarrow \mathbf{u}\cdot\mathbf{v} = 0$. Angle "$= a\pi$": $a = \theta/\pi$ ($\cos\theta = \frac{1}{2} \Rightarrow \frac{1}{3}$; $-\frac{1}{2} \Rightarrow \frac{2}{3}$; $\frac{\sqrt{3}}{2} \Rightarrow \frac{1}{6}$; $\frac{\sqrt{2}}{2} \Rightarrow \frac{1}{4}$).
- $\text{proj}_{\mathbf{v}}\mathbf{u} = \left(\frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{v}|^2}\right)\mathbf{v}$. $\mathbf{u}\times\mathbf{v} = \begin{pmatrix} u_2v_3 - u_3v_2 \\ u_3v_1 - u_1v_3 \\ u_1v_2 - u_2v_1 \end{pmatrix} \perp$ both; $|\mathbf{u}\times\mathbf{v}|$ = parallelogram area; triangle = $\frac{1}{2}|\overrightarrow{AB}\times\overrightarrow{AC}|$; box volume = $|\det[\mathbf{u};\mathbf{v};\mathbf{w}]|$.
- **Line** through P, Q: $\mathbf{r} = P + t(Q-P)$. **Plane**, normal $\mathbf{n} = \langle a,b,c\rangle$ through P_0: $ax+by+cz = \mathbf{n}\cdot P_0$. Three points: $\mathbf{n} = \overrightarrow{PQ}\times\overrightarrow{PR}$.
- **Line ∩ plane**: substitute $x(t), y(t), z(t)$ into the plane, solve t, back-substitute.

SF Q4 $P(3,2,1), Q(0,1,2)$: $(3-3t, 2-t, 1+t)$ in $2x+3y+6z=14$: $18-3t=14$, $t=\frac{4}{3} \Rightarrow \left(-1, \frac{2}{3}, \frac{7}{3}\right)$.

- Distance point–plane: $\frac{|ax_0+by_0+cz_0-d|}{\sqrt{a^2+b^2+c^2}}$; point–line: $\frac{|\overrightarrow{P_0S}\times\mathbf{v}|}{|\mathbf{v}|}$. Planes parallel $\Leftrightarrow$ normals parallel. Line $\parallel$ plane $\Leftrightarrow \mathbf{v}\cdot\mathbf{n} = 0$.

13.3–13.4 Partials & chain rule (Q6–7, 9)

- f_x: differentiate in x, others constant. Clairaut: $f_{xy} = f_{yx}$ — so $f_x = P$, $f_y = Q$ solvable only if $P_y = Q_x$.

Rebuild f from f_x, f_y: $f = \int f_x dx + g(y)$; differentiate in y, match $f_y \Rightarrow g'(y)$; fix the constant from the given value.

SF Q6 $f_x = \cos x + 4x^3y^2$, $f_y = 2x^4y - \sin y$, $f(0,0) = 7$: $f = \sin x + x^4y^2 + g(y)$, $g' = -\sin y \Rightarrow f = \sin x + x^4y^2 + \cos y + 6$.

- **Chain rule = sum over every path**: $z = z(u,v,w)$, each a function of (x,y): $z_y = z_u u_y + z_v v_y + z_w w_y$. An unknown $v(x,y)$ is fine — only v_x, v_y are needed. Substitute u, w back at the end.

SF Q7 $z = u^2 + 2v + w^2$, $u = 3xy$, $v_y = -2y$, $w = x^2 + xy$: $z_y = 2u(3x) + 2(-2y) + 2w(x) = 2x^3 + 20x^2y - 4y$.

- Along a path: $\frac{d}{dt}f(\mathbf{r}(t)) = \nabla f(\mathbf{r}(t))\cdot\mathbf{r}'(t)$ — **evaluate ∇f at the point $\mathbf{r}(t_0)$.**

SF Q9 $\nabla P = \langle yz - 3x^2, xz, xy - 6\rangle$, $\mathbf{r} = \langle t^2, t^3, 2t+1\rangle$, $t=1$: point $(1,1,3)$, $\nabla P = \langle 0,3,-5\rangle$, $\mathbf{r}' = \langle 2,3,2\rangle \Rightarrow -1$.

- Implicit: $F(x,y,z) = 0 \Rightarrow z_x = -F_x/F_z$, $z_y = -F_y/F_z$; $F(x,y) = 0 \Rightarrow y' = -F_x/F_y$.

13.5–13.6 Gradient, directional derivative, tangent planes (Q5, 8)

- $D_{\mathbf{u}}f = \nabla f\cdot\mathbf{u}$ with **u unit**: divide by $|\mathbf{v}|$ ($\langle 3,4\rangle \to 5$, $\langle 5,12\rangle \to 13$, $\langle 8,15\rangle \to 17$). Max rate $|\nabla f|$ along ∇f; min $-|\nabla f|$; zero $\perp \nabla f$.

SF Q8 $f = x^2y - y^2$ at $(3,2)$: $\nabla f = \langle 2xy, x^2 - 2y\rangle = \langle 12, 5\rangle$; along $\langle 8,15\rangle/17$: $(96+75)/17 = \frac{171}{17}$.

- **Tangent plane** to $F(x,y,z) = c$ at P_0: normal $\nabla F(P_0)$, i.e. $\nabla F(P_0)\cdot(\boldsymbol{x} - P_0) = 0$. Graph $z = f(x,y)$: $z = f + f_x(x-a) + f_y(y-b)$. Normal line: $P_0 + t\nabla F$.

Tangent plane parallel to a given plane with normal $\mathbf{n}$: solve $\nabla F = \lambda\mathbf{n}$ for x, y, z in terms of λ (a constant component fixes λ), then use $F = c$ for what remains.

SF Q5 $3x^2 - 2y + 2z^2 = 4$, plane $6x - y + 2z = 5$: $\langle 6x, -2, 4z\rangle = \lambda\langle 6, -1, 2\rangle \Rightarrow \lambda = 2$, $x = 2$, $z = 1$; then $12 - 2y + 2 = 4 \Rightarrow (2,5,1)$.

- Linearization $L = f(a,b) + f_x(x-a) + f_y(y-b)$; $df = f_x dx + f_y dy$.

13.7 Critical points & D-test (Q10)

Solve $f_x = 0 = f_y$ — **factor and split into cases** (e.g. $6xy = 0 \Rightarrow x = 0$ or $y = 0$), never divide by a variable. At each point $D = f_{xx}f_{yy} - f_{xy}^2$:

$D>0$, $f_{xx}<0$	$D>0$, $f_{xx}>0$	$D<0$	$D=0$
local max	local min	saddle	no info

SF Q10 $f = 2x^3 - 6x + 3xy^2$: $f_y = 6xy = 0$. $x = 0 \Rightarrow y = \pm\sqrt{2}$; $y = 0 \Rightarrow x = \pm 1$. $D = 72x^2 - 36y^2$: $\left(0, \pm\sqrt{2}\right)$ **saddles**; $(1,0)$ min; $(-1,0)$ max.

13.8 Lagrange & closed regions (Q11–12)

- Extremise f on $g = c$: solve $\boxed{\nabla f = \lambda\nabla g,\quad g = c}$. Componentwise, $\nabla f = \lambda\nabla g$ reads $f_x = \lambda g_x$, $f_y = \lambda g_y$ (and $f_z = \lambda g_z$) — with $g = c$, these are the "three equations" to write down. Solve by cases, then **evaluate f at every solution**. Two constraints: $\nabla f = \lambda\nabla g + \mu\nabla h$.
- Tricks: factor $(2y(3x-\lambda) = 0)$; eliminate λ ($\nabla f \parallel \nabla g \Leftrightarrow f_x g_y = f_y g_x$); symmetric $\Rightarrow$ try $x = y$; for $x^a y^b z^c$ with $x+y+z = S$: $x : y : z = a : b : c$.

Max/min on a closed disk or region: (1) interior critical points; (2) Lagrange (or parametrise) on the boundary; (3) corners if any. Compare **all** the values — the largest wins.

SF Q11–12 $f = 2x^3 - 6x + 3xy^2$ on $x^2 + y^2 \le 4$. $\nabla f = \lambda\nabla g$: $\langle 6x^2 - 6 + 3y^2, 6xy\rangle = \lambda\langle 2x, 2y\rangle$, with $x^2 + y^2 = 4$. $y = 0$: $f(\pm 2, 0) = \pm 4$. $\lambda = 3x$: $y^2 = 2 = x^2$, $f = \pm 4\sqrt{2}$. Interior max $f(-1,0) = 4 \Rightarrow$ max $4\sqrt{2}$.

- Shortcuts: max of $ax+by+cz$ on a sphere radius R: $R\sqrt{a^2+b^2+c^2}$. Closest point of plane $\mathbf{n}\cdot\boldsymbol{x} = d$ to O: $\left(d/|\mathbf{n}|^2\right)\mathbf{n}$, distance $|d|/|\mathbf{n}|$.

! Answer-format pitfalls

- Fractions in lowest terms; points as (a,b,c); "$a\sqrt{b}$ with b smallest"; "in terms of π"; "integer powers of e" — expand or factor as asked.
- "Rewrite, do not compute" $\Rightarrow$ limits **and** integrand **and** Jacobian, in the requested order of $dudv$ / $d\rho d\varphi d\theta$.
- Normalise directions. Saddle $\Leftrightarrow D < 0$ (sign of f_{xx} irrelevant). Read "saddle points" vs "max value" carefully.

14.1–14.2 Double integrals (Q13–15)

- **Fubini**: $\iint_R f dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f\,dy\,dx$ (vertical strips) $= \int_c^d \int_{h_1(y)}^{h_2(y)} f\,dx\,dy$ (horizontal strips). **Outer limits are constants; inner limits depend only on the outer variable.**
- Rectangle + product: $\iint g(x)h(y)dA = \left(\int g dx\right)\left(\int h dy\right)$.

Choosing the order: draw the region and an arrow through it. If the arrow's **entry** or **exit** curve changes along the region, that order needs two integrals — use the other. Lower limit = where the arrow enters, upper = where it leaves.

SF Q14 Triangle $(1,3),(4,3),(3,4)$: horizontal arrows run from $x = 2y - 5$ to $x = 7 - y$: $\int_3^4 \int_{2y-5}^{7-y} y^2 dx dy = \int_3^4 (12y^2 - 3y^3)dy = \frac{67}{4}$.

SF Q15 Between $y = x^2$ and $y = x + 2$: meet at $x = -1, 2$. $\int_{-1}^2 \int_{x^2}^{x+2} x\,dy\,dx = \int_{-1}^2 (x^2 + 2x - x^3)dx = \frac{9}{4}$.

- Line through $(x_1,y_1),(x_2,y_2)$: $x = x_1 + \frac{x_2 - x_1}{y_2 - y_1}(y - y_1)$. Intersections: equate the curves first.
- **Substitution inside a factor**: $\int_0^a f(kx)dx = \frac{1}{k}\int_0^{ka} f(u)du$.

SF Q17 $\int_0^6 f = 18$, $R = [0,3]\times[0,2]$: $\iint f(2x)f(3y)dA = \left(\frac{18}{2}\right)\left(\frac{18}{3}\right) = 54$.

- **Symmetry**: region symmetric in x and integrand odd in $x \Rightarrow 0$; even $\Rightarrow 2\times$ half.
- Area $= \iint_R 1dA$; volume under $z = f \ge 0$: $\iint f dA$; between surfaces: $\iint \left(z_{\text{top}} - z_{\text{bot}}\right)dA$. Average $= \frac{1}{\text{area}}\iint f$. **Centroid**: $\bar{x} = \frac{1}{A}\iint x dA$ — so $\iint x dA = \bar{x}\cdot A$ (triangle: centroid = mean of vertices).

14.2 Reversing the order (Q16)

Inner integrand has no elementary antiderivative (e^{x^2}, $\sin y/y$, $1/\sqrt{y^4+1}$, $\sqrt{1+y^3}$) $\Rightarrow$ **reverse**.

(1) Read the region off the limits: $a \le x \le b$, $g_1(x) \le y \le g_2(x)$. (2) Sketch it; invert the boundary curves ($y = x^{1/3} \Leftrightarrow x = y^3$). (3) Describe it the other way and rewrite. The new inner integral produces the factor that makes a u-substitution work.

SF Q16 $\int_0^1 \int_{x^{1/3}}^1 \frac{dy\,dx}{\sqrt{y^4+1}}$: region $0 \le y \le 1$, $0 \le x \le y^3$ $\Rightarrow \int_0^1 \frac{y^3}{\sqrt{y^4+1}}dy = \left[\frac{1}{2}\sqrt{y^4+1}\right]_0^1 = \frac{\sqrt{2}-1}{2}$.

Typical flips: $\int_0^a \int_x^a \to \int_0^a \int_0^y$; $\int_0^1 \int_{\sqrt{x}}^1 \to \int_0^1 \int_0^{y^2}$; $\int_0^{2a} \int_{y/2}^a \to \int_0^a \int_0^{2x}$.

14.4 Polar coordinates (Q18–19)

$x = r\cos\theta$, $y = r\sin\theta$, $x^2+y^2 = r^2$, $\boxed{dA = rdrd\theta}$. Use for disks, annuli, sectors, integrands in x^2+y^2.

- Disk radius a: $0 \le r \le a$, $0 \le \theta \le 2\pi$. Line $y = mx$: $\theta = \arctan m$ ($y = x$: $\frac{\pi}{4}$; $y = \sqrt{3}x$: $\frac{\pi}{3}$).
- **Circles through** O: $x^2+y^2 = 2ax \Leftrightarrow r = 2a\cos\theta$, $-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$; $x^2+y^2 = 2ay \Leftrightarrow r = 2a\sin\theta$, $0 \le \theta \le \pi$. ($y = \sqrt{2x-x^2}$ is the upper half of $r = 2\cos\theta$: $0 \le \theta \le \frac{\pi}{2}$.)
- Lines: $x = a \Leftrightarrow r = a\sec\theta$; $y = b \Leftrightarrow r = b\csc\theta$.

SF Q18 R under $y = \sqrt{2x-x^2}$: $\iint \frac{1}{x}dA = \int_0^{\pi/2}\int_0^{2\cos\theta} \frac{1}{r\cos\theta} r dr d\theta = \int_0^{\pi/2} 2d\theta = \pi$.

SF Q19 $\frac{1}{\pi}\iint_{r\le1}\left(3+(x^2+y^2)^2\right)dA = \frac{1}{\pi}\cdot 2\pi\int_0^1(3+r^4)rdr = 2\left(\frac{3}{2}+\frac{1}{6}\right) = \frac{10}{3}$.

- Area in polar: $\int_\alpha^\beta \frac{1}{2}r(\theta)^2 d\theta$. Gaussian: $\int_0^\infty e^{-ax^2}dx = \frac{1}{2}\sqrt{\pi/a}$.
- $\int_0^{\pi/2}\sin^n = \int_0^{\pi/2}\cos^n$: $n=2$: $\frac{\pi}{4}$; 3: $\frac{2}{3}$; 4: $\frac{3\pi}{16}$; 5: $\frac{8}{15}$. Over $[0,\pi]$: double the sin values. $\sin^2\theta = \frac{1-\cos 2\theta}{2}$, $\cos^2\theta = \frac{1+\cos 2\theta}{2}$.

14.8 Change of variables (Q20–22)

$$\iint_R f(x,y)dxdy = \iint_G f(x(u,v),y(u,v))\,|J|\,dudv,$$

$$J = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} = x_u y_v - x_v y_u$$

Recipe. (1) Choose u, v so the boundary curves become $u =$ const, $v =$ const (the region becomes a rectangle). (2) Compute the **easy** Jacobian $\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}$ and use $J = 1/\frac{\partial(u,v)}{\partial(x,y)}$. (3) Write the integrand $\times\,|J|$ in u, v — often the awkward factor cancels. (4) Limits from the boundary curves.

SF Q20 Lines $y = -x+5, -x+2, 2x-1, 2x-4$: $u = x+y \in [2,5]$, $v = y-2x \in [-4,-1]$, $\frac{\partial(u,v)}{\partial(x,y)} = 3$, $x = \frac{u-v}{3}$: $\iint e^x = \frac{1}{3}\int_{-4}^{-1}\int_2^5 e^{u/3}e^{-v/3}dudv = 3e(e-1)^2$.

SF Q21 $u = x^2+y^2 \in [9,16]$, $v = y-2x \in [1,2]$: $\frac{\partial(u,v)}{\partial(x,y)} = 2x+4y$, so $(x+2y)(x^2+y^2)dA = \frac{u}{2}dudv$ $\Rightarrow \frac{1}{4}(256-81) = \frac{175}{4}$.

SF Q22 $u = x^2$, $v = y/x \Rightarrow x = \sqrt{u}$, $y = v\sqrt{u}$, $J = \frac{1}{2}$; $\frac{y^2}{x}e^{x^2} = \sqrt{u}v^2e^u \Rightarrow \int_1^2\int_1^3 \frac{1}{2}u^{1/2}v^2e^u dudv$.

- Common choices: $u = xy$, $v = y/x \Rightarrow \frac{\partial(u,v)}{\partial(x,y)} = 2v$ ($x^2 = u/v$, $y^2 = uv$); $u = x^2-y^2$, $v = xy \Rightarrow 2(x^2+y^2)$; $u = x+y$, $v = x-y$ $\Rightarrow -2$, so $|J| = \frac{1}{2}$.
- Ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2} \le 1$: $x = ar\cos\theta$, $y = br\sin\theta$, $dA = abrdrd\theta$ (area πab). 3-D: $dV = \left|\frac{\partial(x,y,z)}{\partial(u,v,w)}\right| dudvdw$; ellipsoid volume $\frac{4}{3}\pi abc$.

14.5–14.6 Triple integrals (Q23–24)

- $\iiint_D f\,dV$: innermost limits may depend on both outer variables, middle on the outermost only. z first: $z_{\text{bot}(x,y)} \le z \le z_{\text{top}(x,y)}$, then a double integral over the **shadow** in the xy-plane.

SF Q24 $\int_0^1\int_0^x\int_{x^2+y^2}^2 dzdydx = \int_0^1\int_0^x(2-x^2-y^2)dydx = \int_0^1\left(2x-\frac{4}{3}x^3\right)dx = \frac{2}{3}$.

- Box $[a,b]\times[c,d]\times[p,q]$ and product integrand $\Rightarrow$ product of three single integrals.
- **Symmetry (Q23)**: region symmetric under $x \to -x$ and integrand odd in $x \Rightarrow 0$ (e.g. xz^2 on a cube centred at O). By symmetry in a ball: $\iiint x^2 = \iiint y^2 = \iiint z^2 = \frac{1}{3}\iiint \rho^2$.
- Volume $= \iiint 1dV$; mass $M = \iiint \delta dV$; centre of mass $\bar{x} = \frac{1}{M}\iiint x\delta dV$ (etc.); average $= \frac{1}{V}\iiint f$. Tetrahedron with intercepts a, b, c: $V = \frac{abc}{6}$, centroid $\left(\frac{a}{4},\frac{b}{4},\frac{c}{4}\right)$.

14.7 Cylindrical coordinates (Q25–26)

(r,θ,z): $x = r\cos\theta$, $y = r\sin\theta$, $z = z$, $\boxed{dV = rdzdrd\theta}$. Use when the shadow is a disk/annulus or the integrand has x^2+y^2.

- Cylinder $x^2+y^2 = a^2$: $r = a$. Cone $z = \sqrt{x^2+y^2}$: $z = r$. Paraboloid $z = x^2+y^2$: $z = r^2$. Sphere: $z = \pm\sqrt{a^2-r^2}$. Plane $z = 6+x$: $z = 6+r\cos\theta$.
- z **innermost** between the bottom and top surfaces; then r, θ over the shadow. $\int_0^{2\pi}\cos\theta d\theta = \int_0^{2\pi}\sin\theta d\theta = 0$ kills slanted-plane terms.

SF Q25 $1 \le r \le 3$, $0 \le z \le 1$: $\iiint z^2r^4dV = 2\pi\cdot\left[\frac{z^3}{3}\right]_0^1\cdot\left[\frac{r^6}{6}\right]_1^3 = \frac{728\pi}{9}$.

SF Q26 $r \le 3$, $2 \le z \le 6+r\cos\theta$: $\frac{1}{\pi}\int_0^{2\pi}\int_0^3 r^3(4+r\cos\theta)drd\theta = \frac{1}{\pi}\cdot 81\cdot 2\pi = 162$.

- Volume between $z = r^2$ and a plane/paraboloid: equate to get the shadow radius, integrate (top − bottom) $rdrd\theta$.

14.7 Spherical coordinates (Q27–28)

$\rho \ge 0$: distance to O; $\varphi \in [0,\pi]$: angle **down from the** $+z$**-axis**; θ: as in polar.

$$x = \rho\sin\varphi\cos\theta, \quad y = \rho\sin\varphi\sin\theta,$$
$$z = \rho\cos\varphi, \quad r = \rho\sin\varphi$$

$$\boxed{x^2+y^2+z^2 = \rho^2, \quad dV = \rho^2\sin\varphi\, d\rho d\varphi d\theta}$$

Surface	Spherical form
sphere $x^2+y^2+z^2 = a^2$	$\rho = a$
cone $z = \sqrt{x^2+y^2}$	$\varphi = \frac{\pi}{4}$
cone $z = k\sqrt{x^2+y^2}$	$\tan\varphi = 1/k$ ($k=\sqrt{3}$: $\frac{\pi}{6}$; $\frac{1}{\sqrt{3}}$: $\frac{\pi}{3}$)
plane $z = c$	$\rho = c\sec\varphi$
xy-plane / upper half	$\varphi = \frac{\pi}{2}$ / $0 \le \varphi \le \frac{\pi}{2}$
sphere $x^2+y^2+z^2 = 2az$	$\rho = 2a\cos\varphi$, $0 \le \varphi \le \frac{\pi}{2}$
cylinder $x^2+y^2 = a^2$	$\rho = a\csc\varphi$

- Full ball: $0 \le \rho \le a$, $0 \le \varphi \le \pi$, $0 \le \theta \le 2\pi$. First octant: $\varphi, \theta \in \left[0,\frac{\pi}{2}\right]$. $\int_0^\pi \sin\varphi d\varphi = 2$; $\int_0^\alpha \sin\varphi d\varphi = 1-\cos\alpha$.

SF Q27 Unit ball: $\iiint \rho^2 dV = 2\pi\cdot 2\cdot\int_0^1\rho^4 d\rho = \frac{4\pi}{5}$. In general $\iiint_{\rho\le a} g(\rho)dV = 4\pi\int_0^a g(\rho)\rho^2 d\rho$.

SF Q28 $\sqrt{x^2+y^2} \le z \le 2$: cone $\Rightarrow 0 \le \varphi \le \frac{\pi}{4}$; plane $\Rightarrow \rho \le 2\sec\varphi$: $\int_0^{2\pi}\int_0^{\pi/4}\int_0^{2\sec\varphi}\frac{\rho\cos\varphi}{(\rho^2+2)^2}\rho^2\sin\varphi\, d\rho d\varphi d\theta$.

- "Ice-cream cone" (inside $\rho = a$, above the cone $\varphi = \alpha$): $V = \frac{2\pi a^3}{3}(1-\cos\alpha)$. Ball $\frac{4}{3}\pi a^3$; cone $\frac{1}{3}\pi r^2 h$; cylinder $\pi r^2 h$.

+ More worked patterns

shifted disk $\iint_{x^2+y^2\le 2y} ydA = \int_0^\pi\int_0^{2\sin\theta} r^2\sin\theta dr d\theta = \frac{8}{3}\int_0^\pi \sin^4\theta d\theta = \pi$ ($= \bar{y}\cdot$ area).

sector $\int_0^2\int_y^{\sqrt{8-y^2}} \frac{dxdy}{1+x^2+y^2}$: region $0 \le \theta \le \frac{\pi}{4}$, $r \le 2\sqrt{2}$ $\Rightarrow \frac{\pi}{4}\cdot\frac{1}{2}\ln(1+r^2)\Big|_0^{2\sqrt{2}} = \frac{\pi}{4}\ln 3$.

triangle sub $\iint e^{(y-x)/(y+x)}dA$ on the triangle $(0,0),(2,0),(0,2)$: $u = y-x$, $v = y+x$, $|J| = \frac{1}{2}$, region $0 \le v \le 2$, $-v \le u \le v \Rightarrow \frac{1}{2}\int_0^2 v(e-e^{-1})dv = e-\frac{1}{e}$.

volume Between $z = x^2+y^2$ and $z = 2y$: shadow $r \le 2\sin\theta$, $V = \int_0^\pi\int_0^{2\sin\theta}(2r\sin\theta - r^2)rdrd\theta = \frac{4}{3}\int_0^\pi \sin^4\theta d\theta = \frac{\pi}{2}$.

cone $\sqrt{x^2+y^2} \le z \le 3$: $\iiint zdV = 2\pi\int_0^3\int_r^3 zrdzdr = \pi\int_0^3(9r-r^3)dr = \frac{81\pi}{4}$. Or slice: $\int_0^3 z\cdot\pi z^2 dz$.

shifted ball $x^2+y^2+z^2 \le 2z$: $\rho \le 2\cos\varphi$, $\varphi \le \frac{\pi}{2}$: $\iiint \rho dV = 2\pi\int_0^{\pi/2} 4\cos^4\varphi\sin\varphi d\varphi = \frac{8\pi}{5}$.

mass Solid $x^2+y^2 \le z \le 4$, $\delta = z$: slices are disks of area $\pi z \Rightarrow M = \int_0^4 \pi z^2 dz = \frac{64\pi}{3}$, $\bar{z} = \frac{1}{M}\int_0^4 \pi z^3 dz = 3$.

ellipsoid $\frac{x^2}{4}+\frac{y^2}{9}+z^2 \le 1$, $x = 2u$, $y = 3v$, $z = w$: $\iiint z^2dV = 6\iiint_{\text{ball}} w^2 = 6\cdot\frac{4\pi}{15} = \frac{8\pi}{5}$.

! Integral pitfalls

- **Forgetting the Jacobian**: r (polar/cylindrical), $\rho^2\sin\varphi$ (spherical), $|J|$ (substitution, **absolute value**).
- The factor $1/|u_xv_y - u_yv_x|$ comes out in x, y — rewrite it in u, v (or let it cancel) before integrating.
- Outer limits must be numbers. After reversing, re-read **both** limits from the sketch.
- θ-range of $r = 2a\cos\theta$ is $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$, not $[0,2\pi]$; first-quadrant part: $\left[0,\frac{\pi}{2}\right]$.
- φ is measured from the z-axis: the cone $z = r$ is $\varphi = \frac{\pi}{4}$; "above the cone" means $0 \le \varphi \le \frac{\pi}{4}$.
- $\sqrt{x^2+y^2} = r = \rho\sin\varphi$, but $\sqrt{x^2+y^2+z^2} = \rho$. In cylindrical, $x^2+y^2+z^2 = r^2+z^2$.
- Use symmetry **before** computing: odd terms vanish, slanted planes contribute only their constant part.

⇢ Which coordinates? — first move

You see…	Do this
rectangle, polynomial integrand	iterate directly; product ⇒ split
triangle / region between two curves	sketch, pick the single-integral order
inner integral impossible	reverse the order
disk, annulus, x^2+y^2, $\sqrt{2x-x^2}$	polar, $dA = rdrd\theta$
parallelogram; pairs of curves $g = c_1, c_2$	$u = g$, $v = h$; inverse Jacobian
box with odd integrand	symmetry ⇒ 0
cylinder / paraboloid / x^2+y^2 in 3-D	cylindrical, z innermost
ball, cone, $x^2+y^2+z^2$	spherical, $\rho^2\sin\varphi$

∫ Inner integrals that keep appearing

- $\int re^{-r^2}dr = -\frac{1}{2}e^{-r^2}$; $\int \frac{r}{1+r^2}dr = \frac{1}{2}\ln(1+r^2)$; $\int r\sqrt{a^2-r^2}dr = -\frac{1}{3}(a^2-r^2)^{3/2}$.
- $\int\cos^n\varphi\sin\varphi d\varphi = -\frac{\cos^{n+1}\varphi}{n+1}$; $\int\sin^3\varphi d\varphi = -\cos\varphi + \frac{1}{3}\cos^3\varphi$; $\int\sec^2\varphi d\varphi = \tan\varphi$; $\int\tan\varphi d\varphi = \ln|\sec\varphi|$.
- $\int_0^{2\pi}\cos^2\theta d\theta = \int_0^{2\pi}\sin^2\theta d\theta = \pi$; $\int_0^{2\pi}\sin\theta\cos\theta d\theta = 0$.
- Sphere of radius a in cylindrical: volume $= 2\pi\int_0^a 2r\sqrt{a^2-r^2}dr = \frac{4}{3}\pi a^3$ (check).